

Challenge (Habanero)

- Q1.** A recipe for making cookies includes $x^2 + 6x + 8$ cups of flour. Convert this to vertex form.
 (A) $(x + 3)^2 - 1$
 (B) $(x + 4)^2 - 8$
 (C) $(x + 3)^2 + 1$
 (D) $(x + 2)^2 + 4$
 (E) $(x + 1)^2 + 7$
- Q2.** Complete the square to rewrite $x^2 + 4x - 3$.
 (A) $(x + 2)^2 - 7$
 (B) $(x - 2)^2 + 1$
 (C) $(x + 2)^2 - 3$
 (D) $(x - 2)^2 - 7$
 (E) $(x + 3)^2 - 2$
- Q3.** A construction site has a parabolic arch with equation $y = x^2 - 2x - 3$. Convert this to vertex form.
 (A) $y = (x - 1)^2 - 4$
 (B) $y = (x + 1)^2 + 4$
 (C) $y = (x - 2)^2 + 1$
 (D) $y = (x + 2)^2 - 3$
 (E) $y = (x - 1)^2 - 2$
- Q4.** The height of a projectile is given by $y = -x^2 + 4x + 2$. Complete the square to rewrite this equation.
 (A) $y = -(x - 2)^2 + 6$
 (B) $y = -(x + 2)^2 + 2$
 (C) $y = -(x + 4)^2 - 1$
 (D) $y = -(x - 4)^2 + 2$
 (E) $y = -(x - 1)^2 + 3$
- Q5.** A baker is making a cake with $x^2 + 8x + 12$ cups of sugar. Convert this to vertex form.
 (A) $(x + 4)^2 - 4$
 (B) $(x - 2)^2 + 4$
 (C) $(x + 4)^2 + 4$
 (D) $(x - 4)^2 - 12$
 (E) $(x + 6)^2 - 12$
- Q6.** The cost of building a house is given by $y = x^2 + 10x + 5$. Complete the square to rewrite this equation.
 (A) $y = (x + 5)^2 - 20$
 (B) $y = (x - 5)^2 + 20$
 (C) $y = (x + 3)^2 - 4$
 (D) $y = (x + 2)^2 + 1$
 (E) $y = (x - 10)^2 + 5$
- Q7.** A carpenter has a piece of wood with a parabolic shape given by $y = x^2 - 6x - 2$. Convert this to vertex form.
 (A) $y = (x - 3)^2 - 11$
 (B) $y = (x - 2)^2 + 2$
 (C) $y = (x + 2)^2 - 6$
 (D) $y = (x - 4)^2 - 6$
 (E) $y = (x + 6)^2 - 2$
- Q8.** The volume of water in a tank is given by $y = x^2 + 2x - 3$. Complete the square to rewrite this equation.
 (A) $y = (x + 1)^2 - 4$
 (B) $y = (x - 2)^2 - 3$
 (C) $y = (x + 1)^2 + 1$
 (D) $y = (x - 1)^2 + 2$
 (E) $y = (x + 3)^2 - 4$

Open Ended Questions (FRQ)

- Q9.** A construction company is building an arch for a new bridge. The equation of the arch is $y = x^2 - 8x + 7$. Convert this equation to vertex form and interpret the results.
- Q10.** A baker is making a cake with a parabolic shape given by $y = x^2 + 4x + 2$. Complete the square to rewrite this equation and explain the significance of the vertex in this context.